

Griffiths Problem Index in A&A Format

Chapter-by-chapter scaffold for problem statements

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ABSTRACT

Context. This document is a structured manuscript scaffold for the exercises in Griffiths, *Introduction to Electrodynamics*, 4th edition.
Aims. The goal is to organize the textbook exercises by chapter.
Methods. Problem numbers were indexed from the local book PDF and arranged chapter by chapter in a searchable LaTeX source.
Results.
Conclusions.

1. Usage Notes

Problem solving

Problem 1.2. Decide whether the cross product is associative. Either prove that $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ always holds, or give a simple counterexample showing that it does not.

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13 Problem 1.5. Prove the BAC–CAB identity by expanding both sides in Cartesian components.

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16 Problem 1.6. Show that $[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})] + [\mathbf{B} \times (\mathbf{C} \times \mathbf{A})] + [\mathbf{C} \times (\mathbf{A} \times \mathbf{B})] = 0$. Then determine the conditions under which $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$.

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20 Problem 1.7. For a source point at $(2, 8, 7)$ and a field point at $(4, 6, 8)$, compute the separation vector from source to field, its magnitude, and the corresponding unit vector.

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22 Problem 1.8. (a) Show that the two-dimensional rotation matrix preserves dot products. (b) Determine the conditions the entries of a three-dimensional rotation matrix must satisfy in order to preserve the length of every vector.

2. Chapter 1: Vector Analysis

This chapter contains 64 indexed problems.

Problem 1.1. Using the geometric definitions of the dot product and cross product, prove that both operations are distributive. Do this first for three coplanar vectors and then for the general three-dimensional case.

Problem 1.9. Find the rotation matrix corresponding to a 120° rotation about the axis through the origin and the point $(1, 1, 1)$, with the rotation taken clockwise when viewed along the axis toward the origin.

Problem 1.10. Study how vector quantities transform under changes of coordinates. (a) Find how vector components change under a translation. (b) Find how they change under inversion $(x, y, z) \mapsto (-x, -y, -z)$. (c) Determine how a cross product transforms under inversion, decide whether the cross product of two pseudovectors is a vector or pseudovector, and name two classical-mechanics pseudovectors. (d) Determine how the scalar triple product transforms under inversion.

Problem 1.11. Compute the gradient of each of the following scalar fields: (a) $x^2 + y^3 + z^4$, (b) $x^2 y^3 z^4$, and (c) $e^x \sin y \ln z$.

Problem 1.12. The height of a hill is $h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$, where x and y are measured in miles east and north of South Hadley, and h is in feet. Find the hilltop location, the hilltop height, and the slope magnitude and direction at the point one mile east and one mile north of South Hadley.

Problem 1.13. Let $\mathbf{r}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) , and let $r = |\mathbf{r}|$. Show that (a) $\nabla(r^2) = 2\mathbf{r}$, (b) $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$, and (c) derive the corresponding general formula for $\nabla(r^n)$.

Problem 1.14. Let f depend only on y and z . Show that the gradient $\nabla f = (\partial f / \partial y)\hat{\mathbf{y}} + (\partial f / \partial z)\hat{\mathbf{z}}$ transforms as a vector under a rotation in the yz -plane.

Problem 1.15. Compute the divergence of each of the three vector fields $\mathbf{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$, $\mathbf{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3zx\hat{\mathbf{z}}$, and $\mathbf{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}$.

Problem 1.16. Sketch the vector field $\hat{\mathbf{r}}/r^2$ and compute its divergence. Explain why the result may seem surprising at first.

Problem 1.17. Show, in two dimensions, that the divergence transforms as a scalar under rotations.

Problem 1.18. Compute the curl of each of the three vector fields introduced in Problem 1.15.

Problem 1.19. Draw a clockwise tangential vector field around a circle in the xy -plane. By comparing nearby vectors, determine the signs of $\partial v_x / \partial y$ and $\partial v_y / \partial x$, use them to infer the direction of $\nabla \times \mathbf{v}$, and explain how this illustrates the geometric meaning of curl.

Problem 1.20. Construct a nonconstant vector field whose divergence and curl both vanish everywhere.

Problem 1.21. Prove product rules (i), (iv), and (v) from the list given in the text.

Problem 1.22. (a) Write out the meaning of $(\mathbf{A} \cdot \nabla)\mathbf{B}$ in components. (b) Compute $(\hat{\mathbf{r}} \cdot \nabla)\hat{\mathbf{r}}$. (c) For the vector fields $\mathbf{v}_a$ and $\mathbf{v}_b$ from Problem 1.15, evaluate $(\mathbf{v}_a \cdot \nabla)\mathbf{v}_b$.

Problem 1.23. Prove product rules (ii) and (vi), using the component definition of $(\mathbf{A} \cdot \nabla)\mathbf{B}$.

Problem 1.24. Derive the three quotient rules corresponding to the vector-calculus product rules.

Problem 1.25. For $\mathbf{A} = x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3z\hat{\mathbf{z}}$ and $\mathbf{B} = 3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}$, verify product rules (iv), (ii), and (vi) by computing both sides explicitly.

Problem 1.26. Compute the Laplacian of each of the following: (a) $T_a = x^2 + 2xy + 3z + 4$, (b) $T_b = \sin x \sin y \sin z$, (c) $T_c = e^{-5x} \sin 4y \cos 3z$, and (d) the vector field $\mathbf{v} = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$, interpreted componentwise.

Problem 1.27. Prove that the divergence of any curl is identically zero, and check the result explicitly for the vector field $\mathbf{v}_a$ from Problem 1.15.

Problem 1.28. Prove that the curl of any gradient is identically zero, and verify the result for the function in part (b) of Problem 1.11.

Problem 1.29. Evaluate the line integral of $\mathbf{v} = x^2\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + y^2\hat{\mathbf{z}}$ from the origin to $(1, 1, 1)$ along three different paths, and then compute the closed-loop integral obtained by going out along one path and back along another.

Problem 1.30. For the vector field used in Example 1.7, compute the flux through the bottom face of the box, taking upward as positive. Decide whether the surface integral depends only on the boundary curve for this field, and then find the total outward flux through the full closed box.

Problem 1.31. Compute the volume integral of $T = z^2$ over the tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

Problem 1.32. Check the gradient theorem for $T = x^2 + 4xy + 2yz^3$ between $(0, 0, 0)$ and $(1, 1, 1)$ along the three paths shown in Fig. 1.28.

Problem 1.33. Test the divergence theorem for $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$ on the cube of side 2 shown in Fig. 1.30.

Problem 1.34. Test Stokes' theorem for $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$ on the triangular surface shown in Fig. 1.34.

Problem 1.35. Using the same vector field and boundary line as in Example 1.11, check Corollary 1 by integrating over the five faces of the open cube shown in Fig. 1.35.

Problem 1.36. Prove the two integration-by-parts identities given in Eqs. (1.60) and (1.61).

130 **Problem 1.37.** Invert the spherical-coordinate formulas and write r , θ , and ϕ explicitly in terms of x , y , and z .

Problem 1.38. Express the spherical unit vectors $\hat{\mathbf{r}}$, $\hat{\theta}$, and $\hat{\phi}$ in terms of $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, and then invert those relations to express the Cartesian unit vectors in terms of the spherical ones. Check the results using dot and cross products.

Problem 1.39. Check the divergence theorem on a sphere of radius R centered at the origin for (a) $\mathbf{v}_1 = r^2 \hat{\mathbf{r}}$ and (b) $\mathbf{v}_2 = \hat{\mathbf{r}}/r^2$. Compare the two cases and explain any apparent puzzle.

140 **Problem 1.40.** For $\mathbf{v} = (r \cos \theta) \hat{\mathbf{r}} + (r \sin \theta) \hat{\theta} + (r \sin \theta \cos \phi) \hat{\phi}$, compute the divergence and check the divergence theorem on the inverted hemispherical bowl of radius R resting on the xy -plane.

Problem 1.41. For $T = r(\cos \theta + \sin \theta \cos \phi)$, compute the gradient and Laplacian, check the Laplacian by converting to Cartesian coordinates, and verify the gradient theorem along the path shown in Fig. 1.41 from $(0, 0, 0)$ to $(0, 0, 2)$.

Problem 1.42. Express the cylindrical unit vectors $\hat{\mathbf{s}}$, $\hat{\phi}$, and $\hat{\mathbf{z}}$ in terms of the Cartesian basis, and invert those formulas to write $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ in terms of the cylindrical basis.

150 **Problem 1.43.** For $\mathbf{v} = s(2 + \sin^2 \phi) \hat{\mathbf{s}} + s \sin \phi \cos \phi \hat{\phi} + 3z \hat{\mathbf{z}}$: (a) compute the divergence, (b) verify the divergence theorem on the quarter-cylinder shown in Fig. 1.43, and (c) compute the curl.

Problem 1.44. Evaluate the following delta-function integrals: (a) $\int_2^6 (3x^2 - 2x - 1) \delta(x - 3) dx$, (b) $\int_0^5 \cos x \delta(x - \pi) dx$, (c) $\int_0^3 x^3 \delta(x + 1) dx$, and (d) $\int_{-\infty}^{\infty} \ln(x + 3) \delta(x + 2) dx$.

Problem 1.45. Evaluate the following one-dimensional delta-function integrals: (a) $\int_{-2}^2 (2x + 3) \delta(3x) dx$, (b) $\int_0^2 (x^3 + 3x + 2) \delta(1 - x) dx$, (c) $\int_{-1}^1 9x^2 \delta(3x + 1) dx$, and (d) $\int_{-\infty}^a \delta(x - b) dx$.

160 **Problem 1.46.** (a) Show that $x d\delta(x)/dx = -\delta(x)$. (b) Let $\theta(x)$ be the step function. Show that $d\theta/dx = \delta(x)$.

Problem 1.47. Write the charge density in delta-function form for (a) a point charge q located at $\mathbf{r}'$, (b) an electric dipole made of $-q$ at the origin and $+q$ at $\mathbf{a}$, and (c) a uniformly charged infinitesimally thin spherical shell of radius R and total charge Q centered at the origin.

Problem 1.48. Evaluate the following three-dimensional delta-function integrals: (a) $\int (r^2 + \mathbf{r} \cdot \mathbf{a} + a^2) \delta^3(\mathbf{r} - \mathbf{a}) d\tau$, (b) $\int_V |\mathbf{r} - \mathbf{b}|^2 \delta^3(5\mathbf{r}) d\tau$ over the cube of side 2 centered at the origin, (c) $\int_V [r^4 + r^2(\mathbf{r} \cdot \mathbf{c}) + c^4] \delta^3(\mathbf{r} - \mathbf{c}) d\tau$ over a sphere of radius 6, and (d) $\int_V \mathbf{r} \cdot (\mathbf{d} - \mathbf{r}) \delta^3(\mathbf{e} - \mathbf{r}) d\tau$ over the specified shifted sphere. 170

Problem 1.49. Evaluate $J = \int_V e^{-r} \nabla \cdot (\hat{\mathbf{r}}/r^2) d\tau$ over a sphere of radius R centered at the origin, using two different methods as in Example 1.16.

Problem 1.50. (a) For $\mathbf{F}_1 = x^2 \hat{\mathbf{z}}$ and $\mathbf{F}_2 = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$, compute the divergence and curl of each. Determine which can be written as a gradient field and which can be written as a curl field, and find corresponding scalar or vector potentials. (b) Show that $\mathbf{F}_3 = yz\hat{\mathbf{x}} + zx\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$ can be represented both ways, and find both potentials.

Problem 1.51. For Theorem 1, prove the chain of implications $(d) \Rightarrow (a) \Rightarrow (c) \Rightarrow (b) \Rightarrow (c) \Rightarrow (a)$. 180

Problem 1.52. For Theorem 2, prove the chain of implications $(d) \Rightarrow (a) \Rightarrow (c) \Rightarrow (b) \Rightarrow (c) \Rightarrow (a)$.

Problem 1.53. Using the three vector fields from Problem 1.15, determine which can be written as gradients of scalar functions and find such scalars. Also determine which can be written as curls of vector potentials and construct appropriate vector potentials.

Problem 1.54. Check the divergence theorem for $\mathbf{v} = r^2 \cos \theta \hat{\mathbf{r}} + r^2 \cos \phi \hat{\theta} - r^2 \cos \theta \sin \phi \hat{\phi}$ on one octant of a sphere of radius R , making sure to include every piece of the boundary surface. 190

Problem 1.55. Check Stokes' theorem for $\mathbf{v} = ay\hat{\mathbf{x}} + bx\hat{\mathbf{y}}$, where a and b are constants, around a circle of radius R in the xy -plane centered at the origin.

Problem 1.56. Compute the line integral of $\mathbf{v} = 6\hat{\mathbf{x}} + yz^2\hat{\mathbf{y}} + (3y + z)\hat{\mathbf{z}}$ around the triangular path shown in Fig. 1.49, and verify the result independently with Stokes' theorem.

Problem 1.57. Compute the closed line integral of $\mathbf{v} = (r \cos^2 \theta) \hat{\mathbf{r}} - (r \cos \theta \sin \theta) \hat{\theta} + 3r \hat{\phi}$ around the path in Fig. 1.50, working in either spherical or cylindrical coordinates, and check the result using Stokes' theorem. 200

Problem 1.58. Verify Stokes' theorem for the vector field $\mathbf{v} = y\hat{\mathbf{z}}$ on the triangular surface shown in Fig. 1.51.

Problem 1.59. Check the divergence theorem for $\mathbf{v} = r^2 \sin \theta \hat{\mathbf{r}} + 4r^2 \cos \theta \hat{\boldsymbol{\theta}} + r^2 \tan \theta \hat{\boldsymbol{\phi}}$ over the "ice-cream cone" volume shown in Fig. 1.52, whose cap is a spherical surface of radius R .

210 **Problem 1.60.** Use the integral theorems to recover known identities for second derivatives. (a) Combine the gradient theorem with Stokes' theorem for $\mathbf{v} = \nabla T$. (b) Combine Stokes' theorem with the divergence theorem in the analogous way.

Problem 1.61. Derive the listed corollaries of the gradient, divergence, and curl theorems: (a) $\int_V \nabla T d\tau = \int_S T d\mathbf{a}$; (b) $\int_V (\nabla \times \mathbf{v}) d\tau = - \int_S \mathbf{v} \times d\mathbf{a}$; (c) Green's identity; (d) Green's second identity; and (e) $\int_S \nabla T \times d\mathbf{a} = - \oint_P T d\mathbf{l}$.

220 **Problem 1.62.** The vector area of a surface is $\mathbf{a} = \int_S d\mathbf{a}$. (a) Find the vector area of a hemispherical bowl of radius R . (b) Show that every closed surface has zero vector area. (c) Show that surfaces with the same boundary have the same vector area. (d) Prove $\mathbf{a} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{l}$. (e) Show that $\oint (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l} = \mathbf{a} \times \mathbf{c}$ for constant $\mathbf{c}$.

Problem 1.63. For the field $r^n \hat{\mathbf{r}}$: (a) compute its divergence directly, test the result with the divergence theorem, decide whether a delta function is needed at the origin, and derive the general formula for $\nabla \cdot (r^n \hat{\mathbf{r}})$; (b) compute its curl and verify the conclusion using Problem 1.61(b).

230 **Problem 1.64.** Define $D(r, \epsilon) = -(1/4\pi) \nabla^2 [1/\sqrt{r^2 + \epsilon^2}]$. Show that as $\epsilon \rightarrow 0$, this quantity approaches $\delta^3(\mathbf{r})$ by proving: (a) an explicit formula for $D(r, \epsilon)$, (b) $D(0, \epsilon) \rightarrow \infty$, (c) $D(r, \epsilon) \rightarrow 0$ for $r \neq 0$, and (d) the integral of $D(r, \epsilon)$ over all space is always 1.

3. Chapter 2: Electrostatics

This chapter contains 61 indexed problems.

240 **Problem 2.1.** Place twelve equal charges q at the vertices of a regular 12-gon and a test charge Q at the center. (a) Find the net force on Q . (b) Remove the charge at "6 o'clock" and find the new force on Q , explaining the symmetry argument carefully. (c) Repeat the central-force question for 13 equal charges on a regular 13-gon. (d) Remove one of those 13 charges and determine the resulting force on Q .

Problem 2.2. Find the electric field at a point a height z above the midpoint of two equal and opposite point charges $\pm q$ separated by a distance d . Give both magnitude and direction.

Problem 2.3. A finite straight segment of length L carries uniform line charge density λ . Find the electric field at a point a height z above one end of the segment, and check that the result behaves sensibly in the limit $z \gg L$.

Problem 2.4. Find the electric field at a point a height z above the center of a square loop of side a carrying uniform line charge density λ .

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Problem 2.5. Find the electric field at a point a height z above the center of a circular loop of radius r carrying uniform line charge density λ .

Problem 2.6. Find the electric field at a point a height z above the center of a uniformly charged disk of radius R and surface charge density σ . Check the limits $R \rightarrow \infty$ and $z \gg R$.

Problem 2.7. A spherical shell of radius R carries uniform surface charge density σ . Compute the electric field both inside and outside the shell, express the answer in terms of the total charge q , and handle carefully the cases $z < R$ and $z > R$.

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Problem 2.8. Use the result for the charged spherical shell to find the electric field inside and outside a uniformly charged solid sphere of radius R and total charge q . Then sketch $|\mathbf{E}|$ as a function of distance from the center.

Problem 2.9. Suppose the electric field is $\mathbf{E} = kr^3 \hat{\mathbf{r}}$ in spherical coordinates. (a) Find the corresponding charge density ρ . (b) Compute the total charge enclosed in a sphere of radius R in two different ways.

Problem 2.10. A point charge q is placed at a back corner of a cube. Find the electric flux through the shaded face of the cube.

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Problem 2.11. Use Gauss's law to find the electric field inside and outside a spherical shell of radius R carrying uniform surface charge density σ , and compare the result with the direct integration of Problem 2.7.

Problem 2.12. Use Gauss's law to determine the electric field inside a uniformly charged solid sphere of volume charge density ρ , and compare the answer with the result of Problem 2.8.

Problem 2.13. Use Gauss's law to find the electric field at distance s from an infinitely long straight wire carrying uniform line charge density λ .

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Problem 2.14. Find the electric field inside a sphere whose charge density is $\rho = kr$, where k is a constant.

Problem 2.15. A thick spherical shell occupies the region $a \leq r \leq b$ and has charge density $\rho = k/r^2$. Find the electric field in the three regions $r < a$, $a < r < b$, and $r > b$. Then plot $|\mathbf{E}|$ versus r for the special case $b = 2a$.

Problem 2.16. A long coaxial cable has an inner solid cylinder of radius a carrying uniform volume charge density ρ , and an outer cylindrical shell of radius b carrying just enough negative surface charge density to make the cable electrically neutral. Find the electric field in the regions $s < a$, $a < s < b$, and $s > b$, and sketch $|\mathbf{E}|$ as a function of s .

Problem 2.17. An infinite slab of thickness $2d$ carries uniform volume charge density ρ . Taking $y = 0$ at the midplane, find the electric field as a function of y and plot the signed field component $E_y(y)$.

Problem 2.18. Two overlapping spheres of equal radius R carry uniform charge densities $+\rho$ and $-\rho$. Let $\mathbf{d}$ point from the positive center to the negative center. Show that the electric field in the region of overlap is uniform and determine its value.

Problem 2.19. Compute $\nabla \times \mathbf{E}$ directly from Coulomb's-law integral expression for $\mathbf{E}$, following the strategy used earlier for the divergence.

Problem 2.20. One of the two given vector fields is impossible as an electrostatic field. Identify which one cannot arise from static charges. For the one that is possible, find a potential V using the origin as reference and verify the result by taking $-\nabla V$.

Problem 2.21. Find the electrostatic potential inside and outside a uniformly charged solid sphere of radius R and total charge q , using infinity as the zero of potential. Differentiate your answer to recover the field, and sketch $V(r)$.

Problem 2.22. Find the potential at distance s from an infinitely long straight wire carrying uniform line charge density λ . Then compute its gradient and verify that it reproduces the correct electric field.

Problem 2.23. For the thick spherical shell of Problem 2.15, compute the electric potential at the center, taking infinity as the reference point.

Problem 2.24. For the coaxial cable of Problem 2.16, find the potential difference between a point on the axis and a point on the outer cylinder.

Problem 2.25. Using the standard potential formulas for point, line, and surface charge distributions, find the potential at height z above the center of the three charge distributions shown in

Fig. 2.34. In each case compute $\mathbf{E} = -\nabla V$ and compare with earlier field results. Then change one of the two equal point charges in case (a) to $-q$, infer what the potential suggests about the field, and explain carefully why that suggestion disagrees with the actual field from Problem 2.2.

Problem 2.26. A conical surface of height h and top radius h carries uniform surface charge density σ . Find the potential difference between the vertex of the cone and the center of the circular top.

Problem 2.27. Find the potential on the axis of a uniformly charged solid cylinder of length L , radius R , and volume charge density ρ , at a point a distance z from the center. Then use the result to calculate the electric field on the axis for $z > L/2$.

Problem 2.28. Use the general potential integral to compute the potential inside a uniformly charged solid sphere of radius R and total charge q , and compare the answer with Problem 2.21.

Problem 2.29. Verify directly that the Coulomb-potential integral formula satisfies Poisson's equation by applying the Laplacian and using the delta-function identity for $\nabla^2(1/r)$.

Problem 2.30. (a) Check that the electric fields found in Examples 2.5 and 2.6 and in Problem 2.11 satisfy the boundary condition for the normal component across a surface charge. (b) Use Gauss's law to find the field inside and outside a long hollow cylindrical tube carrying uniform surface charge density σ , and compare with the same boundary condition. (c) Check that the result of Example 2.8 is consistent with the relevant boundary conditions on $\mathbf{E}$ and V .

Problem 2.31. Three charges occupy three corners of a square of side a , as shown in Fig. 2.41. (a) How much work is required to bring in a fourth charge $+q$ from infinity and place it at the remaining corner? (b) How much work is required to assemble the entire four-charge configuration from infinity?

Problem 2.32. Two positive point charges q_A and q_B , with masses m_A and m_B , are held at rest a distance a apart by a massless string. The string is cut and the charges fly apart. Find the speed of each charge when they are very far from one another.

Problem 2.33. An infinite one-dimensional lattice of alternating charges $\pm q$ is arranged along the x -axis, with spacing a between nearest neighbors. Find the work per particle needed to assemble the chain and determine the corresponding dimensionless Madelung constant.

Problem 2.34. Find the electrostatic energy stored in a uniformly charged solid sphere of radius R and total charge q in three different ways: (a) by integrating $\rho V/2$, (b) by integrating the field-energy density over all space, and (c) by using

370 the integration-by-parts form involving both volume and surface terms.

Problem 2.35. Recompute the energy of a uniformly charged solid sphere by building it up layer by layer: at each stage bring in an infinitesimal charge from infinity and spread it uniformly over the surface of the growing sphere. Determine dW for an increase dr in radius and integrate to the final radius R .

380 Problem 2.36. Two concentric spherical shells of radii a and b carry charges $+q$ and $-q$, uniformly distributed. Compute the electrostatic energy of the system (a) from the field-energy formula and (b) from the capacitance formula using the potential difference between the shells.

Problem 2.37. Evaluate the interaction-energy term $\epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau$ for two point charges q_1 and q_2 separated by distance a .

390 Problem 2.38. A conducting sphere of radius R carrying charge q is surrounded by a thick concentric conducting shell with inner radius a and outer radius b , the shell having zero net charge. (a) Find the surface charge densities on the three spherical surfaces $r = R$, $r = a$, and $r = b$. (b) Find the potential at the center, with infinity as reference. (c) Now ground the outer surface and determine how the answers to parts (a) and (b) change.

Problem 2.39. Inside a neutral conducting sphere of outer radius R , two spherical cavities of radii a and b contain point charges q_a and q_b at their centers. (a) Find the surface charge densities on the two cavity walls and on the outside surface. (b) Find the field outside the conductor. (c) Find the field in each cavity. (d) Find the forces on q_a and q_b . (e) State which answers would change if a third external charge were brought near the conductor.

400 Problem 2.40. (a) If a point charge q sits inside a cavity in an uncharged conductor, is the force on q necessarily zero? (b) Is the force between a point charge and a nearby uncharged conductor always attractive?

Problem 2.41. Two large conducting plates of area A are separated by a small distance d , and each carries charge Q . Find the electrostatic pressure between the plates.

Problem 2.42. A conducting sphere of radius R carries total charge Q . Find the net force of repulsion exerted by the southern hemisphere on the northern hemisphere.

410 Problem 2.43. Find the capacitance per unit length of two coaxial conducting cylinders of radii a and b .

Problem 2.44. The plates of a parallel-plate capacitor move closer together by a small distance ϵ under their mutual attraction. (a) Use the field-pressure formula to compute the work done by electrostatic forces in terms of the field E and plate area A . (b) Use the field-energy expression to compute the corresponding decrease in stored energy.

Problem 2.45. Find the electric field at height z above the center of a square sheet of side a carrying uniform surface charge density σ . Check the limits $a \rightarrow \infty$ and $z \gg a$.

Problem 2.46. A vector field in spherical coordinates is given 420 by $\mathbf{E}(r, \theta, \phi) = k[r^3 \hat{\mathbf{r}} + 2 \sin \theta \cos \theta \sin \phi \hat{\boldsymbol{\theta}} + \sin \theta \cos \phi \hat{\boldsymbol{\phi}}]$. Find the charge density ρ that produces it.

Problem 2.47. Find the net force exerted by the southern hemisphere of a uniformly charged solid sphere on the northern hemisphere, expressing the result in terms of the sphere radius R and total charge Q .

Problem 2.48. An inverted hemispherical bowl of radius R carries uniform surface charge density σ . Find the potential difference between the north pole and the center of the hemisphere.

Problem 2.49. A sphere of radius R has volume charge density 430 $\rho(r) = kr$. Find the total electrostatic energy of the configuration, and verify the result by at least two different methods.

Problem 2.50. The potential of some charge distribution is $V(r) = Ae^{-\lambda r}/r$. Determine the electric field, the charge density, and the total charge of the configuration.

Problem 2.51. Find the potential at a point on the rim of a uniformly charged disk of radius R and surface charge density σ . First reduce the answer to a dimensionless integral for a constant k , and then evaluate k analytically if possible or numerically if necessary. 440

Problem 2.52. Two infinitely long wires parallel to the x -axis carry line charges $+\lambda$ and $-\lambda$. (a) Find the potential at any point (x, y, z) , taking the origin as reference. (b) Show that the equipotentials are circular cylinders and determine the axis and radius corresponding to a given potential V_0 .

Problem 2.53. Analyze the space-charge-limited planar vacuum diode. Electrons start at a cathode held at zero potential and accelerate toward an anode at potential V_0 . Assuming the plate area is large compared with the gap, so all quantities depend only on x : (a) write Poisson's equation, (b) relate electron speed 450 to $V(x)$, (c) use steady current to relate ρ and v , (d) eliminate ρ and v to get a differential equation for V , (e) solve for $V(x)$, $\rho(x)$, and $v(x)$, and compare $V(x)$ to the empty-gap potential, and (f) derive the Child–Langmuir law $I = KV_0^{3/2}$ and determine K .

Problem 2.54. Assume Coulomb's law is corrected so that the force between point charges becomes $\mathbf{F} = (1/4\pi\epsilon_0) q_1 q_2 e^{-r/\lambda} (1 + r/\lambda) \hat{\mathbf{r}}/r^2$, with λ a huge new length scale. Reformulate electrostatics under this law, assuming superposition still holds: (a) write the field of a continuous charge distribution, (b) determine whether a scalar potential still exists and justify your answer, (c) find the point-charge potential using infinity as reference, (d) derive the modified integral law for a sphere centered on a point charge, (e) generalize it to arbitrary charge distributions, (f) redraw the usual ρ - $\mathbf{E}$ - V relation diagram for this modified theory, and (g) show that charge on a conductor now splits between the volume and the surface.

Problem 2.55. Suppose the electric field is $\mathbf{E} = (ax, 0, 0)$, where a is constant. Find the charge density, and explain how the field can point everywhere in the same Cartesian direction even though the charge density is uniform.

Problem 2.56. Construct the gravitational analog of electrostatics for a uniform sphere of mass M and radius R . Compute its gravitational self-energy, estimate the gravitational energy of the Sun from observed parameters, and compare the implied lifetime from gravitational radiation power alone with the actual age of the Sun.

Problem 2.57. Use the known surface-charge formula for a charged ellipsoid to generate three limiting conductor problems by suitable choices of a , b , and c : (a) the net surface charge density on a circular conducting disk of radius R , (b) the net surface charge density on an infinite conducting ribbon extending from $x = -a$ to $x = +a$, and (c) the charge per unit length on a conducting needle running from $x = -a$ to $x = +a$. Sketch each resulting distribution.

Problem 2.58. (a) Three equal point charges q sit at the vertices of an equilateral triangle inscribed in a circle of radius a . Besides the center, there are three other points inside the triangle where the field vanishes. Locate them numerically. (b) For a regular n -gon of equal charges, there are n additional interior zero-field points. Find their radial positions for $n = 4$ and $n = 5$, and discuss what seems to happen as $n \rightarrow \infty$.

Problem 2.59. Decide whether the following theorem is true: if a charged conductor experiences force $\mathbf{F}$ in an external field $\mathbf{E}_e$, then reversing the external field to $-\mathbf{E}_e$ reverses the force to $-\mathbf{F}$. Either prove the claim or give a counterexample. Then reconsider the question under the additional assumption that the external field is uniform.

Problem 2.60. A point charge q sits at the center of an uncharged spherical conducting shell with inner radius a and outer radius b . How much work is required to remove the charge to infinity through a tiny hole in the shell?

Problem 2.61. For N equal point charges constrained to lie on or inside a circle of radius R , determine the minimum-energy

configuration. In particular, show numerically that for $N = 12$ the lowest-energy state has 11 charges on the boundary and one at the center, and decide whether the $N = 11$ minimum has all 11 on the boundary or 10 on the boundary plus one at the center. Express energies in units of $q^2/(4\pi\epsilon_0 R)$.

Problem 2.61. *Problem statement intentionally omitted here. Add a paraphrase or licensed transcription if needed.*

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4. Chapter 3: Potentials

This chapter contains 58 indexed problems.

Problem 3.1. Compute the average electrostatic potential over a spherical surface of radius R produced by a point charge q located inside that sphere. Then show the more general formula $V_{\text{ave}} = V_{\text{center}} + Q_{\text{enc}}/(4\pi\epsilon_0 R)$, where V_{center} is the contribution at the center from all charges outside the sphere and Q_{enc} is the total enclosed charge.

Problem 3.2. Give a one-sentence justification of Earnshaw's theorem: a charged particle cannot be held in stable equilibrium by electrostatic forces alone. Then apply that reasoning to the cube of fixed charges discussed in the chapter and explain where the apparent "electrostatic bottle" fails.

520

Problem 3.3. Solve Laplace's equation in spherical coordinates for the special case $V = V(r)$ only. Do the analogous calculation in cylindrical coordinates for the case $V = V(s)$ only.

Problem 3.4. (a) Show that the average electric field over a spherical surface due to charges outside the sphere equals the field at the center. (b) Determine the corresponding average field due to charges located inside the sphere.

530

Problem 3.5. Prove that the electrostatic field is uniquely determined when the charge density ρ is specified in the volume and, on each boundary surface, either the potential V or the normal derivative $\partial V/\partial n$ is given. Do not assume the boundaries are conductors or equipotentials.

Problem 3.6. Re-derive the second uniqueness theorem using Green's identity, taking $T = U = V_3$, and fill in the missing steps of the proof.

Problem 3.7. A grounded conducting plane lies in the xy -plane. Two point charges are above it on the same vertical line: a charge $+q$ at height $3d$ and a charge $-2q$ at height d . Find the net force on the charge $+q$.

540

Problem 3.8. For the image solution of a point charge outside a grounded conducting sphere: (a) rewrite the potential in spherical coordinates using the law of cosines, (b) find the induced surface charge density $\sigma(\theta)$ and integrate it to obtain the total

induced charge, and (c) compute the electrostatic energy of the configuration.

550 **Problem 3.9.** Modify the image-charge construction for a conducting sphere that is held at a fixed nonzero potential V_0 rather than grounded. Determine what additional image charge is needed and where it must be placed. Then use the result to find the force between a point charge q and a neutral conducting sphere.

Problem 3.10. An infinite straight wire carrying uniform line charge density λ runs parallel to the x -axis at height d above a grounded conducting plane. (a) Find the potential in the region above the plane. (b) Determine the induced surface charge density on the conductor.

560 **Problem 3.11.** Two grounded conducting half-planes meet at right angles. A point charge q is placed in the wedge between them. Construct the necessary image charges, find the potential in the physical region, compute the force on q , and determine the work required to bring the charge in from infinity. Then discuss for which wedge angles the method of images still works.

Problem 3.12. Two infinitely long conducting cylinders of radius R have centers separated by distance $2d$. One cylinder is held at potential $+V_0$ and the other at $-V_0$. Find the electrostatic potential everywhere.

570 **Problem 3.13.** In the infinite slot problem of Example 3.3, replace the boundary at $x = 0$ by two conducting strips: $0 \leq y \leq a/2$ is kept at $+V_0$, while $a/2 \leq y \leq a$ is kept at $-V_0$. Find the potential inside the slot.

Problem 3.14. For the infinite slot of Example 3.3 with the strip at $x = 0$ held at constant potential V_0 , determine the induced charge density $\sigma(y)$ on that strip.

580 **Problem 3.15.** A rectangular pipe extends infinitely along the z -direction. The sides $y = 0$, $y = a$, and $x = 0$ are grounded, while the side $x = b$ is held at a prescribed potential $V_0(y)$. (a) Derive a general formula for the potential inside the pipe. (b) Specialize to the case $V_0(y) = V_0$ constant.

Problem 3.16. A cube of side a has five grounded faces; the sixth face is insulated from the others and maintained at constant potential V_0 . Find the potential inside the cube. Also infer, by symmetry or superposition, what the potential at the center should be and check that your formula is consistent with it.

590 **Problem 3.17.** Use Rodrigues' formula to derive $P_3(x)$. Verify directly that $P_3(\cos \theta)$ satisfies the angular differential equation for $l = 3$, and check by explicit integration that P_3 and P_1 are orthogonal.

Problem 3.18. (a) Suppose the surface of a sphere is held at constant potential V_0 . Use the spherical separation-of-variables machinery to find the potential inside and outside the sphere. (b) Using the same method, find the potential inside and outside a spherical shell carrying uniform surface charge density σ_0 .

Problem 3.19. A sphere of radius R has prescribed surface potential $V(R, \theta) = k \cos(3\theta)$, with no charge present either inside or outside. Find the potential in both regions and determine the resulting surface charge density $\sigma(\theta)$.

Problem 3.20. Let the potential $V_0(\theta)$ be specified on the surface of a sphere, with no free charge inside or outside. Show that the surface charge density can be written as a Legendre-series expression in terms of the expansion coefficients of $V_0(\theta)$. 600

Problem 3.21. A conducting sphere of radius R carries total charge Q and is placed in an otherwise uniform external field $\mathbf{E}_0$. Find the potential outside the sphere and state clearly what reference level you are using for the potential.

Problem 3.22. Starting from the known on-axis potential of a uniformly charged disk, derive the first three terms of the Legendre expansion for the off-axis potential when $r > R$. Then obtain the interior expansion for $r < R$, treating the upper and lower hemispherical regions separately. 610

Problem 3.23. A spherical shell of radius R carries surface charge $+\sigma_0$ on the northern hemisphere and $-\sigma_0$ on the southern hemisphere. Find the potentials inside and outside the shell, working out the expansion coefficients explicitly through A_6 and B_6 .

Problem 3.24. Solve Laplace's equation in cylindrical coordinates by separation of variables under the assumption of no z -dependence. Be sure to include all radial solutions, including the one needed to represent the potential of an infinite line charge. 620

Problem 3.25. An infinitely long conducting cylinder of radius R is placed in a uniform electric field $\mathbf{E}_0$ perpendicular to its axis. Using the cylindrical solutions from Problem 3.24, find the external potential and the induced surface charge density.

Problem 3.26. A surface charge density $\sigma(\phi) = a \sin(5\phi)$ is fixed on an infinite cylindrical shell of radius R . Determine the electrostatic potential inside and outside the cylinder.

Problem 3.27. A sphere of radius R centered at the origin carries the axially symmetric volume charge density $\rho(r, \theta) = k r^2(R - 2r) \sin \theta$, where k is a constant. Find the far-field approximation to the potential on the z -axis. 630

Problem 3.28. A circular ring of radius R in the xy -plane carries uniform line charge density λ . Find the first three terms, $n = 0, 1, 2$, in the multipole expansion of the potential $V(r, \theta)$.

Problem 3.29. Four point charges are each a distance a from the origin: $3q$ lies on the $+z$ -axis, q lies on the $+y$ -axis, and two charges $-2q$ lie on the $\pm x$ -axis. Find a simple far-field approximation for the potential, expressed in spherical coordinates.

640 **Problem 3.30.** For the spherical shell with surface charge density $\sigma = k \cos \theta$: (a) compute the dipole moment of the shell, and (b) compare the exact potential with the far-field dipole approximation to determine whether any higher multipole moments are present.

Problem 3.31. For the physical dipole of Example 3.10, expand $1/r_{\pm}$ through order $(d/r)^3$ and use the result to identify the quadrupole and octopole contributions to the potential.

650 **Problem 3.32.** Two point charges, $3q$ and $-q$, are separated by distance a . For each of the three arrangements shown in Fig. 3.35, compute the monopole moment, the dipole moment, and the large- r potential including both the monopole and dipole terms.

Problem 3.33. An ideal dipole $\mathbf{p} = p\hat{\mathbf{z}}$ is fixed at the origin. Find the force on a point charge q placed at (a) $(a, 0, 0)$ and (b) $(0, 0, a)$. Then determine the work required to move the charge from $(a, 0, 0)$ to $(0, 0, a)$.

660 **Problem 3.34.** Three point charges lie at distance a from the origin in the arrangement of Fig. 3.38. Find the far-field electric field in spherical coordinates, keeping the two lowest nonvanishing orders in the multipole expansion.

Problem 3.35. A solid sphere of radius R is centered at the origin. The northern hemisphere carries uniform volume charge density $+\rho_0$, and the southern hemisphere carries $-\rho_0$. Find the approximate field $\mathbf{E}(r, \theta)$ for $r \gg R$.

Problem 3.36. Show that the electric field of an ideal dipole can be written in coordinate-free form as $\mathbf{E}_{\text{dip}}(\mathbf{r}) = (1/4\pi\epsilon_0 r^3) [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$.

670 **Problem 3.37.** Give an alternative proof of the spherical averaging theorem using only Laplace's equation. Let $V_{\text{ave}}(R)$ be the average of the potential over a sphere of radius R centered at a point P , derive dV_{ave}/dR in terms of a flux integral, and use the divergence theorem to show that $V_{\text{ave}}(R) = V(P)$ in a charge-free region.

Problem 3.38. Re-derive the induced surface charge density on a grounded conducting plane due to a point charge q at height d , but do so without the method of images. Write the normal field just below the conductor as the sum of the field from q and the field from the unknown surface charge density $\sigma(x, y)$, set the total to zero, and solve for σ .

Problem 3.39. Two infinite grounded conducting planes are parallel and separated by distance a . A point charge q lies between them, at distance x from one plate. Find the force on the charge and check the limits $a \rightarrow \infty$ and $x = a/2$. 680

Problem 3.40. Two infinitely long straight wires carrying line charges $+\lambda$ and $-\lambda$ lie on opposite sides of a long conducting cylinder of radius R . The wires are each at distance a from the cylinder axis, and the cylinder carries no net charge. Find the potential everywhere.

Problem 3.41. Approximate the C_{60}^- ion as a conducting spherical shell of radius $R = 3.5 \text{ \AA}$ carrying one extra electron smeared uniformly over its surface. A second electron experiences attraction at short distance but repulsion far away. (a) Find numerically the radius r where the net force on the second electron vanishes. (b) Find the energy required to bring an electron from infinity to that radius. 690

Problem 3.42. Use superposition of separated solutions to find the potential in a rectangular pipe for which the two faces $x = \pm b$ are held at V_0 , the face $y = a$ is held at V_1 , and the face $y = 0$ is grounded.

Problem 3.43. A conducting sphere of radius a , held at potential V_0 , is surrounded by a thin concentric spherical shell of radius b carrying surface charge density $\sigma(\theta) = k \cos \theta$. (a) Find the potential in the regions $r > b$ and $a < r < b$. (b) Determine the induced surface charge density on the inner conductor. (c) Find the total charge of the system and check that it matches the large- r behavior of the potential. 700

Problem 3.44. A total charge $+Q$ is distributed uniformly along the segment of the z -axis from $-a$ to $+a$. Derive the multipole expansion of the potential for $r > a$, showing that only even Legendre polynomials appear and obtaining the first few terms explicitly. 710

Problem 3.45. A long cylindrical shell of radius R carries uniform surface charge $+\sigma_0$ on its upper half and $-\sigma_0$ on its lower half. Find the electrostatic potential both inside and outside the cylinder.

Problem 3.46. A thin insulating rod runs along the z -axis from $z = -a$ to $z = +a$. In each of the cases $\lambda(z) = k \cos(\pi z/2a)$, $\lambda(z) = k \sin(\pi z/a)$, and $\lambda(z) = k \cos(\pi z/a)$, determine the leading nonvanishing term in the multipole expansion of the potential.

720 **Problem 3.47.** Show that the average electric field over the volume of a sphere of radius R , due to the charge contained inside that sphere, is $\mathbf{E}_{\text{ave}} = -\mathbf{p}/(4\pi\epsilon_0 R^3)$, where $\mathbf{p}$ is the total dipole moment of the enclosed charge distribution. Follow the suggested route: first analyze a single interior point charge by comparison with a uniformly charged sphere, then generalize by superposition, and also show that the average field from exterior charges equals the field at the center.

730 **Problem 3.48.** (a) Compute the volume average of the dipole field over a sphere centered at the dipole and compare it with the general averaging theorem. (b) Use that comparison to infer the singular field contribution at the origin, thereby recovering the delta-function term in the exact field of an ideal dipole.

Problem 3.49. For the spherical shell with $\sigma(\theta) = k \cos \theta$, take the limit $R \rightarrow 0$ and show that the result reproduces the delta-function term in the exact dipole field.

740 **Problem 3.50.** (a) Prove Green's reciprocity theorem: if ρ_1 produces V_1 and ρ_2 produces V_2 , then $\int \rho_1 V_2 d\tau = \int \rho_2 V_1 d\tau$. Use two evaluations of $\int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau$ and integration by parts. (b) Apply reciprocity to two separated conductors to show that if charge Q on conductor a produces potential V_{ab} on conductor b , then the same charge on b produces the same potential $V_{ba} = V_{ab}$ on a .

Problem 3.51. Use Green's reciprocity theorem to solve two induced-charge problems. (a) A point charge q lies between the grounded plates of a parallel-plate capacitor; find the total induced charge on each plate in terms of the plate spacing d and the charge position x . (b) A point charge q lies between two grounded concentric conducting spheres of radii a and b ; find the total induced charge on each sphere.

750 **Problem 3.52.** (a) Rewrite the quadrupole contribution to the multipole expansion in tensor form and define the quadrupole moment tensor Q_{ij} . (b) For the square of side a in the xy -plane centered at the origin, with corner charges $-q, +q, +q, -q$ arranged as in Fig. 3.30, compute all nine components of Q_{ij} . (c) Show that the quadrupole moment is independent of the choice of origin when both the monopole and dipole moments vanish. (d) Define the octopole moment and write the corresponding octopole term in the multipole expansion.

760 **Problem 3.53.** Re-solve the problem of a conducting sphere in a uniform external electric field by the method of images, and verify that the result agrees with the solution previously obtained by separation of variables.

Problem 3.54. In the infinite rectangular pipe, let the bottom $y = 0$ and the side walls $x = \pm b$ be grounded, while the top $y = a$ is held at constant potential V_0 . Find the potential inside. Set up the separation in the style of Example 3.4, include the $k = 0$ mode, and work out the full solution.

Problem 3.55. (a) A long square metal pipe of side a is grounded on three sides, while the fourth side is insulated and kept at constant potential V_0 . Find the net charge per unit length induced on the opposite side. (b) Repeat the question for a circular pipe of radius R split into four equal longitudinal conducting sections, three grounded and one held at V_0 ; find the charge per unit length on the opposite quarter. 770

Problem 3.56. An ideal electric dipole is fixed at the origin, pointing along the z -axis. A point charge is released from rest somewhere in the xy -plane. Show that its subsequent trajectory is a semicircle, like the motion of a pendulum whose support point is at the origin.

Problem 3.57. A positive point charge q of mass m moves in a circular orbit of radius s in the field of a fixed dipole $\mathbf{p} = p\hat{\mathbf{z}}$ at the origin. Determine the orientation of the orbital plane and compute the speed, angular momentum, and total energy of the orbiting charge. 780

Problem 3.58. Find the surface charge density $\sigma(\theta)$ on a sphere of radius R that reproduces, for all exterior points, the same field as a point charge q located on the z -axis at position $a < R$.

5. Chapter 4: Electric Fields in Matter

This chapter contains 43 indexed problems.

Problem 4.1. A hydrogen atom, with Bohr radius about $0.5 \text{ \AA}$, sits between two parallel metal plates separated by 1 mm and connected to a 500 V source. Estimate what fraction of the atomic radius the induced displacement d represents, and then estimate the voltage required to ionize the atom with that same apparatus. Use the polarizability listed in Table 4.1. 790

Problem 4.2. For a hydrogen atom in its ground state, the electron cloud has charge density $\rho(r) = q(\pi a^3)^{-1} e^{-2r/a}$, where q is the electron charge and a is the Bohr radius. Compute the atomic polarizability. A suggested route is to first determine the field of the electron cloud and then expand for $r \ll a$.

Problem 4.3. Equation 4.1 says that an induced dipole moment is proportional to the applied field. Test that rule for a model atom whose electron-cloud density grows in proportion to distance from the center up to radius R . Determine the power of E that appears in $p(E)$, and find the condition on $\rho(r)$ that guarantees a linear weak-field response. 800

Problem 4.4. A point charge q is placed far from a neutral atom with polarizability α . Find the attractive force between the charge and the atom.

Problem 4.5. Two perfect dipoles, $\mathbf{p}_1$ and $\mathbf{p}_2$, are separated by distance r in the geometry shown in Fig. 4.6. Find the torque on $\mathbf{p}_1$ due to $\mathbf{p}_2$, and the torque on $\mathbf{p}_2$ due to $\mathbf{p}_1$, each taken about that dipole's own center. 810

Problem 4.6. A perfect dipole $\mathbf{p}$ is located a height z above an infinite grounded conducting plane. The dipole makes angle θ with the normal to the plane. Find the torque on the dipole and determine the orientation of stable equilibrium if it is free to rotate.

Problem 4.7. Show that the potential energy of an ideal dipole in an external electric field is $U = -\mathbf{p} \cdot \mathbf{E}$.

820 **Problem 4.8.** Derive the interaction energy of two dipoles separated by displacement $\mathbf{r}$, expressing the answer in terms of $\mathbf{p}_1$, $\mathbf{p}_2$, and $\hat{\mathbf{r}}$. Use the dipole-energy formula together with the electric field of a dipole.

Problem 4.9. A dipole $\mathbf{p}$ is located a distance r from a point charge q , and $\mathbf{p}$ makes angle θ with the vector from q to the dipole. Find (a) the force on the dipole and (b) the force on the point charge.

830 **Problem 4.10.** A sphere of radius R has polarization $\mathbf{P}(\mathbf{r}) = k\mathbf{r}$, where k is constant. Compute the bound surface and volume charge densities, and then determine the electric field inside and outside the sphere.

Problem 4.11. A short cylinder of radius a and length L carries a permanent uniform polarization parallel to its axis. Determine the bound charge distribution and sketch the field lines for the limiting cases $L \gg a$, $L \ll a$, and $L \sim a$.

Problem 4.12. For the uniformly polarized sphere of Example 4.2, compute the potential directly from Eq. 4.9 rather than from the bound-charge picture.

840 **Problem 4.13.** An infinitely long cylinder of radius a carries a uniform polarization $\mathbf{P}$ perpendicular to its axis. Find the electric field inside the cylinder, and show that the exterior field can be written in the form given in the text in terms of a , s , and $\mathbf{P}$.

Problem 4.14. Show directly from the formulas for σ_b and ρ_b that the total bound charge of any polarized neutral dielectric is zero.

850 **Problem 4.15.** A dielectric spherical shell with inner radius a and outer radius b has frozen-in polarization $\mathbf{P}(\mathbf{r}) = (k/r)\hat{\mathbf{r}}$. There is no free charge. Find the electric field in all three regions: first by locating the bound charges and using Gauss's law for $\mathbf{E}$, and then by computing $\mathbf{D}$ first and recovering $\mathbf{E}$.

Problem 4.16. Inside a large dielectric, the field is $\mathbf{E}_0$ and the displacement is $\mathbf{D}_0 = \epsilon_0\mathbf{E}_0 + \mathbf{P}$. A small cavity is carved out while the polarization remains frozen. Find the field and displacement at the center of (a) a spherical cavity, (b) a long needle-shaped

cavity parallel to $\mathbf{P}$, and (c) a thin wafer-shaped cavity perpendicular to $\mathbf{P}$. Assume the cavity is small enough that $\mathbf{P}$, $\mathbf{E}_0$, and $\mathbf{D}_0$ are essentially uniform.

Problem 4.17. For the bar electret of Problem 4.11, with $L \approx 2a$, draw careful field-line sketches for $\mathbf{P}$, $\mathbf{E}$, and $\mathbf{D}$. Use the distinction that $\mathbf{E}$ lines begin and end on total charge, while $\mathbf{D}$ lines 860 begin and end only on free charge.

Problem 4.18. A parallel-plate capacitor is filled by two dielectric slabs, each of thickness a , so the plate separation is $2a$. The upper slab has dielectric constant 2, the lower slab has dielectric constant 1.5, and the free surface charge densities on the plates are $\pm\sigma$. Find $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{P}$ in each slab, the potential difference, the full bound-charge distribution, and then recompute the fields from all charges as a check.

Problem 4.19. You have just enough linear dielectric of dielectric constant ϵ_r to occupy half of a parallel-plate capacitor. 870 Compare two arrangements: one where the dielectric fills half the plate area, and one where it fills half the plate separation. For each arrangement, determine the fractional change in capacitance. Then, for a specified plate voltage V , find $\mathbf{E}$, $\mathbf{D}$, $\mathbf{P}$, and all free and bound surface charges in every region.

Problem 4.20. A solid sphere of radius R is made of linear dielectric material and contains a uniform free charge density ρ . Find the electrostatic potential at the center, measured relative to infinity, if the dielectric constant is ϵ_r .

Problem 4.21. A coaxial cable has inner conductor radius a 880 and outer conductor inner radius c . The annular region from b to c is filled with dielectric of constant ϵ_r , while the region from a to b is empty. Find the capacitance per unit length.

Problem 4.22. A very long cylinder of linear dielectric material, radius a and susceptibility χ_e , is placed in an otherwise uniform electric field $\mathbf{E}_0$ perpendicular to its axis. Determine the electric field inside the cylinder.

Problem 4.23. Re-derive the internal field of a dielectric sphere placed in a uniform external field by using successive approximations. Start with $\mathbf{E}_0$, compute the induced polarization $\mathbf{P}_0$, 890 then the field it creates, then the next correction to the polarization, and so on. Sum the resulting series and compare with Eq. 4.49.

Problem 4.24. An uncharged conducting sphere of radius a is coated with an insulating spherical shell of dielectric constant ϵ_r out to radius b . The coated sphere is placed in a uniform external electric field $\mathbf{E}_0$. Find the electric field throughout the insulating layer.

Problem 4.25. In the dielectric-interface problem of Example 4.8, suppose the upper half-space is also filled with a linear dielectric, but with susceptibility χ'_e different from that below the plane. Determine the electrostatic potential in all space.

Problem 4.26. A conducting sphere of radius a carries charge Q and is surrounded by linear dielectric material of susceptibility χ_e out to radius b . Use Eq. 4.58 to compute the energy stored in the configuration.

Problem 4.27. For a sphere of radius R with frozen uniform polarization $\mathbf{P}$, compute the energy once from Eq. 4.55 and once from Eq. 4.58. Explain why the two answers differ and discuss which one, if either, should be interpreted as the physical energy of the system.

Problem 4.28. Two long coaxial metal cylinders stand vertically in dielectric oil of susceptibility χ_e and mass density ρ . The inner cylinder is held at potential V , and the outer cylinder is grounded. Determine how high the oil rises between the cylinders.

Problem 4.29. Return to the two-dipole arrangement of Problem 4.5. Compute the force on $\mathbf{p}_2$ from $\mathbf{p}_1$ and the force on $\mathbf{p}_1$ from $\mathbf{p}_2$, and comment on Newton's third law. Then find the total torque on $\mathbf{p}_2$ about the center of $\mathbf{p}_1$ and compare it with the torque on $\mathbf{p}_1$ about that same point.

Problem 4.30. An electric dipole directed along the y -axis is placed midway between two large conducting plates that make equal small angles θ with the x -axis and are held at potentials $+V$ and $-V$. Without doing a long calculation, determine the direction of the net force on the dipole and justify your reasoning qualitatively.

Problem 4.31. A fixed point charge Q sits at the center of a circular frictionless track of radius R . A dipole $\mathbf{p}$ rides on the track and is constrained to remain tangent to the circle. Use the dipole-force formula to show that the force is always forward along the track, and then explain why this does not create a perpetual-motion machine.

Problem 4.32. Earnshaw's theorem forbids electrostatic trapping of a charged particle. Investigate whether a neutral but polarizable atom can be trapped instead. First show that $\mathbf{F} = \frac{1}{2}\alpha\nabla(E^2)$. Then show that E^2 cannot have a local maximum in a charge-free region, so no stable electrostatic trap is possible.

Problem 4.33. A dielectric cube of side a , centered at the origin, has frozen-in polarization $\mathbf{P} = k\mathbf{r}$. Find all bound charges and verify that their total is zero.

Problem 4.34. A parallel-plate capacitor is filled with dielectric whose dielectric constant varies linearly from 1 at the lower plate

$x = 0$ to 2 at the upper plate $x = d$. The plates are connected to a battery of voltage V . Determine the complete bound-charge distribution and check that the net bound charge vanishes.

Problem 4.35. A point charge q is embedded at the center of a linear dielectric sphere of radius R and susceptibility χ_e . Find the electric field, the polarization, the bound volume charge density, and the bound surface charge density. Also determine the total bound charge on the surface and identify where the compensating negative bound charge is located.

Problem 4.36. At the boundary between two linear dielectrics with no free surface charge, show that the field-line angles satisfy $\tan\theta_2/\tan\theta_1 = \epsilon_2/\epsilon_1$. Then discuss whether a convex dielectric lens would tend to focus or defocus the electric field.

Problem 4.37. A point dipole $\mathbf{p}$ is embedded at the center of a dielectric sphere of radius R and dielectric constant ϵ_r . Find the electrostatic potential both inside and outside the sphere.

Problem 4.38. Prove the uniqueness theorem for a region containing a specified free-charge distribution together with linear dielectrics of known susceptibilities. Show that if the potential is specified on the boundary, then the potential everywhere inside is unique. Use the suggested integral involving $\nabla \cdot (V_3\mathbf{D}_3)$.

Problem 4.39. A conducting sphere at potential V_0 is half buried in linear dielectric occupying the half-space $z < 0$. Test the claim that the potential is exactly the same as if the dielectric were absent. Do this by writing the candidate potential, finding the resulting field, polarization, bound charge, and free surface charge, showing that these reproduce the same potential, and then invoking the uniqueness theorem. Finally discuss whether the same idea works for the two modified geometries shown in Fig. 4.36.

Problem 4.40. Use the force formula $\mathbf{F} = \int (\mathbf{P} \cdot \nabla) \mathbf{E}_{\text{ext}} d\tau$ to find the force on a small dielectric sphere of radius R and susceptibility χ_e located a distance s from a thin wire carrying uniform line charge density λ .

Problem 4.41. Relate microscopic polarizability α to macroscopic susceptibility χ_e . Starting from the naive estimate $P = N\alpha E$, account for the difference between the macroscopic field E and the local field acting on a given atom, E_{else} . Assuming each atom occupies a spherical cell of radius R , derive the Clausius-Mossotti formula connecting α , N , and ϵ_r .

Problem 4.42. Check the Clausius-Mossotti relation numerically for the gases listed in Table 4.1, using dielectric constants from Table 4.2. Comment on why the correction term is hard to distinguish from the naive low-density approximation in gases.

Problem 4.43. Derive the Langevin theory for a polar dielectric. First, using the Boltzmann distribution for dipole orientations in an external field, obtain the polarization law $P = Np[\cosh(pE/kT)/\sinh(pE/kT) - kT/(pE)]$ and sketch $P/(Np)$ as a function of pE/kT . Then take the weak-field, high-temperature limit to extract the susceptibility, and compare the prediction with data for liquid water at 20°C and for water vapor at 100°C and 1 atm.

6. Chapter 5: Magnetostatics

This chapter contains 62 indexed problems.

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7. Chapter 6: Magnetic Fields in Matter

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9. Chapter 8: Conservation Laws

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10. Chapter 9: Electromagnetic Waves

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11. Chapter 10: Potentials and Fields

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12. Chapter 11: Radiation

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13. Chapter 12: Electrodynamics and Relativity

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